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Exploring research ethics through the lens of critical posthumanism in the age of Artificial Intelligence

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ABSTRACT

Drawing on critical posthumanism, I attempt to theorize an alternative, non-linear conceptualization of graduate students' experiences with AI ethics in the context of an educational research methodology course. The conceptual tools of assemblage, distributive agency and ethical criticality were used as the theoretical framework to examine the complex, relational processes that graduate students ought to engage in to develop AI ethics. As the advent of AI has added further complexity to research ethics, my reflections focus on three key tensions explored in this conceptual paper: (1) blurring lines of authorship, (2) relinquishing agency and (3) abating criticality. The emergent pedagogy adopted in the teaching of this course was useful in responding to the ethical challenges that graduate students, as novice researchers, face in the age of AI. The paper concludes with a discussion on how to better support graduate students in developing AI ethics through pedagogy and policy.

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1. Introduction

Recent advances in Artificial Intelligence (AI), particularly those pertaining to text generation, have provided a wealth of new tools and services, almost suddenly becoming readily accessible to university students (Chaaban et al. 2024; Schlagwein and Willcocks 2023). Not long after the global urgency to shift all instruction online, ChatGPT – a leading text generation agent – took Higher Education (HE) by storm, and students and faculty, not yet recovered from educational disruption, were to encounter another major change. Several hundreds of AI agents followed suite, promising researchers automated data analysis and interpretation, literature reviews and resource allocation, among other functions (AlAfnan et al. 2023). Amidst excitement and bafflement, AI was touted as an invaluable tool for teaching, learning and research processes in HE (Jang, Choi, and Kim 2022; Van Dis et al. 2023). Despite these promises, the challenges to research ethics have been daunting, causing heightened concerns over student plagiarism and academic misconduct (Hill-Yardin et al. 2023; Robin 2023).

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Beyond their extraordinary capabilities to produce texts in response to any prompt, AI systems ‘hallucinate’ plausible looking responses, which are not necessarily accurate (AlAfnan et al. 2023; Schlagwein and Willcocks 2023). Their responses are based on patterns of likeness, or associations between words and concepts, having been trained on large datasets (Schlagwein and Willcocks 2023). These advanced capabilities are met with what has become known as the ‘black box’ problem, making it difficult to understand and interpret how complex AI algorithms arrive at their outputs, opening ethical debates pertaining to transparency and accountability, bias and fairness, and informed consent. These concerns, among others, have urged several non-profit organizations (e.g. UNESCO 2023), government representatives, and HE institutions to develop ethical codes of conduct in abundance (Adams et al. 2022; Jang, Choi, and Kim 2022). Journal editors were also compelled to engage in the ethical debate; most advocating for a cautious teleological assessment of AI use, ‘where it is a useful tool for advancing knowledge’ (Schlagwein and Willcocks 2023, 235). This teleological assessment of ethical use is based on the premise that AI enhances productivity rates by ‘employing the best tools and technologies at hand’ (Schlagwein and Willcocks 2023, 235).

However, such usefulness of AI may also be contested on ethical grounds. It may be argued that ‘a useful tool for advancing knowledge’ now has the capacity of doing so in unethical ways (AlAfnan et al. 2023; Kramm and McKenna 2023). Therefore, while offering efficiency, some have argued that the overreliance on AI can shift the focus from nurturing critical engagement with knowledge to merely producing automated results that can evade detection (Kramm and McKenna 2023), thus undermining genuine learning and the transformative nature of education for graduate students. This brings to the fore a deontological assessment of AI, as advocated by the likes of Emanuel Kant, where it is the process that counts, not primarily the outcome (Schlagwein and Willcocks 2023). Traditionally, ethical assessments in education have often reflected this Kantian sense of duty that is translated into academic integrity protocols and codes of conduct, which may very well be considered universally accepted principles across many HE institutions. These protocols are meant to set standards for accepted ethical student behaviors, such as honesty, fairness and responsibility, against which students are held accountable.

Critiques of this approach contend that adherence to fixed principles overlooks the fluid context-dependent nature of ethical decision-making (Frisicaró-Pawlowski 2018; Vetter et al. 2024) and refute the expectation that students will act out of a sense of duty and comply willingly with predetermined ethical codes. For instance, Smith (2016) has argued for understanding ethical decision-making as dynamic, deeply intertwined within specific rhetorical and disciplinary contexts. This approach suggests that ethical sensibility should be adaptive and responsive to the unique demands of each discursive situation, rather than anchored in static codes of conduct. In alignment with this view, Vetter et al. (2024) call for an ecological understanding of AI ethics and introduces the concept of a ‘local ethics’ (2). This framework encourages the negotiations of AI ethics at the ground level between students, AI agents and institutional codes, and so positions the classroom as the space for ethical exploration. Similarly, Frisicaró-Pawlowski (2018) advocates for a dialogical process for the teaching of ethics, one that evolves within local contexts and is subject to continuous negotiation. This shift mirrors broader theoretical advancements in AI ethics, where the emphasis moves away from indoctrinating students with predetermined values toward facilitating their ability to adapt ethically across different contexts (Vetter et al. 2024).

Situated within the larger discussions on HE pedagogy and policy, this conceptual paper aims to uncover some of the challenges to graduate students' experiences with AI ethics that may result in their non-compliance with mandated codes of conduct. What is needed is a non-linear understanding of the ways that students learn to use AI ethically and put such learning into their daily interactions with these agents. Critical posthumanism (Braidotti 2013, 2019) is one theory that offers conceptual tools to develop this view in a way that opens possibilities to consider different elements within systems, as well as their intra-actions (Barad 2007). Notwithstanding concerns pertaining to the application of theories to novel situations in mechanistic ways, I engage with critical posthumanism as a flexible framework that offers some useful tools to explore the way graduate students, as novice researchers, AI agents and ethical codes co-constitute one another, while avoiding an overly deterministic application of this theory.

Constituting the context of this study, my reflections are grounded within an educational research methodology course which I taught using an emergent pedagogy; one that is flexible, adaptive and responsive to students' needs and allows the course content to evolve organically in real-time based on their intra-actions, feedback and engagements with AI ethics. By exploring these intra-actions, I attempt to uncover the tensions and challenges graduate students face in developing AI ethics. This conceptual paper concludes with a thoughtful discussion on ways to better support graduate students in developing critical awareness of their entanglements with AI, by introducing them to possibilities for agentic participation and ethical engagement in an AI-saturated world.

2. Educational research with AI

AI has revolutionized research activities, creating both opportunities and challenges. By embracing generative AI, researchers have entered an era characterized by accelerated innovation, reduced publication timelines and enhanced diversity in scientific discourse (Van Dis et al. 2023). There is compelling reason for graduate students to embrace AI as they write up literature reviews or analyze qualitative datasets under time constraints. The potential of this technology can also be seen in providing feedback to improve student writing and reasoning skills (Van Dis et al. 2023). Researchers tend to agree that AI will have transformational impact on the way research is produced and disseminated (Coffey 2023; Conroy 2023). The results of a survey indicating that researchers currently using AI are still a minority (Van Noorden and Perkel 2023) may already be outdated, due to the rapid changes occurring in the deployment and use of AI in research activities (Foltynek et al. 2023).

It has also been argued that AI addresses certain social injustices, such as the advantage of being conversant with dominant cultural and linguistic abilities, where previously those who did not possess this valued capital were left out. AI can assist individuals in articulating their thoughts with greater fluency and has the potential to democratize knowledge. It is believed to mitigate language barriers and improve accessibility for non-native English speakers or those less experienced in academic writing, leading to a more inclusive research environment and fostering a broader range of perspectives (AlAfnan et al. 2023; Vetter et al. 2024). Hwang et al. (2023) assert the benefits of AI-assisted research, specifically for non-native English researchers, alleviating their need for English editing services and enabling them to dedicate more attention to domain-specific inquiries.

Despite the apparent advantage of speedy productions, AI is also said to be at the ‘peak of inflated (hype) expectations’ (Schlagwein and Willcocks 2023, 232), as humans struggle with ethical and practical limitations. For instance, AI could degrade the quality and transparency of research and compete with human agency (Van Dis et al. 2023; Vetter et al. 2024). Schlagwein and Willcocks (2023) rightly caution against the long-term, hidden and non-obvious negative implications: ‘New, powerful technologies set us on new epistemological, political and social trajectories. They are not merely additive but ecological – they have the potential to reshape everything’ (237). Similarly, Van Dis et al. (2023) and Foltynnek et al. (2023) also caution against the increase in inaccuracies and biases in publications, the difficulties in detecting research outputs produced by AI and the lack of transparency, among other limitations.

3. Critical posthumanism: an introduction

Critical posthumanism represents a diverse spectrum of theoretical perspectives, analytical approaches, philosophies and critiques (Taylor and Bayley 2019). This varied collection of ideas, despite its internal variations, shares a common goal: To challenge the traditional humanist worldview, marked by its emphasis on hierarchical structures, binary oppositions, linear progression, cause and effect and a rationalist approach. It tends to categorize entities into oppositional and value-ranked pairs, such as nature versus culture, or teacher versus student, positioning the concept of the White, Eurocentric male as the standard for defining humanity (Braidotti 2013). Over the years, humanism has become a ‘commonsense’ ideology, forming the backbone of prevailing social, political and cultural norms (Braidotti 2013). In response, critical posthumanism aims to eradicate these familiar dichotomies and substitute these binaries with hybrid assemblages and relational networks, as will be discussed in the next section.

Notably, posthumanism theories, such as those proposed by Barad (2007) and Morton (2013) are rooted in physics, serving the foundation for broader assertions about ecological and sociopolitical phenomena, and raising questions related to truth (e.g. How can students tell if a text generated by ChatGPT is true or reliable?) and reality (e.g. How can students tell if a source is a deepfake?). Posthumanism emerged as a successor of postmodernity with its focus on deconstructing radical anthropocentric worldviews that place humans at the top of a hierarchical system (Nath and Manna 2023). This ideology strongly opposes the notion that humans deserve preference over nature or other life forms simply because they are the most rational species (Nath and Manna 2023). With technological progress accelerating since the late twentieth century, this perspective is becoming more widely accepted among scholars.

Against this backdrop, and to apply critical posthumanism in a way that yields deeper understandings of students’ use of AI in research activities, this conceptual paper attends to the main tenets of critical posthumanism explained through three central ideas: assemblages, distributive agency and ethical criticality.

3.1. Assemblages

Critical posthumanism marks a departure from traditional humanist views where the individual human is seen as the primary focus of reality and social interactions. In

contrast, a posthuman perspective shifts attention to the concept of the assemblage (Deleuze and Guattari 1987). An assemblage is understood as a complex, interconnected network that incorporates a wide range of elements – social, discursive, cultural, psychological, historical, emotional and material (Strom and Martin 2022). These elements come together in a fluid, temporary formation to produce something new, highlighting the intricate, relational, and dynamic essence of reality and the interconnectedness of all phenomena (Leander and Burriss 2020). Accordingly, phenomena come into existence through the workings of the assemblage, through their relationality and the intra-actions among their elements. According to Barad (2007), the ‘primary ontological units are not ‘things’ but phenomena – dynamic topological reconfigurings/entanglements/ relationalities/(re)articulations of the world’ (141).

Using these ideas, graduate students’ research activities become entangled within an assemblage encompassing human participants (e.g. students, instructors), material objects (e.g. AI agents, computers), discursive elements (e.g. educational content, ethical guidelines), as well as the environment in which the intra-actions among these elements occur (e.g. online platforms, classroom settings). Additionally, this assemblage would also integrate the wider discourse around ethical AI use (e.g. HE policies on AI use, public attitudes towards AI), as well as broader socio-cultural, economic and regulatory elements (e.g. data privacy laws, access to digital resources, global trends regarding AI). This assemblage forms a dynamic network where each component influences and intra-acts with the others, shaping how students engage with AI in an (un)ethical way. This ontological view, which emphasizes dynamism over thingness, encourages students to think of themselves alongside AI, entangled in a dynamic network that changes how they think of their identity and their agency. Beyond a description of AI as merely a tool that students use (Adams et al. 2022), AI feeds on their discursive intra-actions and shapes their research activities in often subtle and unconscious ways. Such complexity poses challenges to the universality of ethical standards and linearity of AI checklists. The assumption that students will unarguably adhere to ethical guidelines has been problematized (Smith 2016). By extension, they also challenge assessment measures, which can no longer be reduced to plagiarism percentages derived from tools that fail to detect AI (Kramm and McKenna 2023) or to an instructor’s hunch toward a student’s potential cheating with AI (Robin 2023).

3.2. Distributive agency

Another defining concept within critical posthumanism is distributive agency, which is also a characteristic of assemblages (Bennett 2010). This concept challenges the view of individuals as autonomous, powerful and self-regulating agents (Miller 2016). Instead, it starts from the position that they share agency with all elements of an assemblage, both human and nonhuman (Leander and Burriss 2020). Distributive agency shifts the focus away from the human agent and emphasizes the interconnected and dynamic interactions among all elements in the assemblage (Adams et al. 2022). This approach suggests that agency is a collective capacity that emerges and evolves through the continuous interplay of social and material agents. Critical posthumanism further postulates that intentionality is not a pre-requisite of agency (Leander and Burriss 2020). On this point, Latour (2005) argues that ‘if action is limited a priori to what ‘intentional’,

‘meaningful’ humans do, it is hard to see how a hammer, a basket, a door closer, a cat, a rug, a mug, a list, or a tag could act’ (71). He asserts that while these objects do not ‘cause’ or ‘determine what will happen in a course of action’, yet they should not be relegated to serving as ‘mere intermediaries’, as they may ‘authorize, allow, afford, encourage, permit, suggest, influence, block, render possible, and so on’ (72). Adams et al. (2022) challenge the notion of AI as ‘simply a tool taken up for instrumental purpose’, rather AI tools ‘participate in co-constituting complex human-technology hybrids that mobilise new actions, transform knowledge frameworks and inaugurate novel ways of being in the world’ (3).

In the context of graduate students’ research activities, the advent of AI-powered text generators has introduced a partnership model, where students and AI tools jointly produce written material (Leander and Burriss 2020). In this collaborative, the responsibility and capability for writing are dispersed among humans and technological objects alike, leading to a mutually enhancing dynamic (Eaton 2023). This view refutes dominant paradigms of agency that postulate humans as using AI tools as objects, and considers them as agentive actors, challenging traditional ideas about authorship and revealing the potential of technology to act as a co-agent in the writing of text. Thus, the result is a ‘product that is neither fully written by a human, nor by an AI, but one that is hybrid’ (Eaton 2023, 3). Attempting to determine where the human ends and where the AI begins is an exercise in futility (Eaton 2023). This perspective prompts thinking about how algorithms, data practices and user interfaces shape ethical behavior in tandem with human agency.

This depiction of distributive agency is by no means a demotion of the human capacity to influence their actions or make ethical decisions. Human agents have played and continue to play a role in the development, deployment and control of algorithmically agentive outcomes. The fact that text generative tools have the capacity to spill out results does not absolve humans of their responsibilities toward verifying, interpreting or rejecting AI outputs (Schlagwein and Willcocks 2023). Texts generated by AI are by no means static and can be revised edited and reworked (Eaton 2023). It is in such fluidity, as well as in the sophistication of AI outputs that students learn to escape detection software (Eaton 2023). Students can enact agency by retaining control over what they write or accept the agency of the AI tools if they choose (Eaton 2023). In both cases, they remain responsible for what is written, and obliged to use fact-checking and verification procedures to ensure the validity, accuracy and reliability of their research outputs (Van Dis et al. 2023). Scholarship, according to Schlagwein and Willcocks (2023), ‘inherently embraces moral agency’ (236), and prods researchers to declare their AI use in as much the same way that they would declare the use of any other tool or technique.

3.3. Ethical criticality

As noted in the introduction, traditional ethical systems in deontology (rules-based ethics) and teleology (consequence-oriented ethics) have typically relied on ‘discrete actions and identifiable agents and outcomes’ (Johnstone 2007, 74). Yet, the advent of AI has complicated these concepts and negated their relevance to a ‘worlding’ with AI (Barad 2007, 160). As AI continues to evolve in a networked relation with the human,

ethical considerations require a renewed theoretical stance including ethico-onto-epistemological possibilities (Barad 2007). According to Barad (2007, 160), ‘ethics is not simply about responsible actions in relation to human experiences of the world; rather, it is a question of material entanglements and how each intra-action matters in the reconfiguring of these entanglements, that is, it is a matter of the ethical call that is embodied in the very worlding of the world’ (Barad 2007, 160).

Through this theoretical stance, ethics is redefined as an ongoing process of engaging with the socio-material world, where every intra-action contributes to the reconfiguration of an assemblage. In these entanglements, AI tools are not merely computational agents that ‘shape us and we them (and even challenge this dichotomy of human/computational agent), but rather a world where human agents can leverage computational machines and processes to become more ethical assemblages with them’ (Leander and Burris 2020, 1274). Thus, in agreement with Luke (2018), ‘the educational challenge raised by digital culture is not one of skill or technological competence, but one of participation and ethics’ (185). Beyond the traditional deontological/teleological dichotomy, work in the ethics of AI becomes especially helpful for developing ethical criticality along with these complex computational systems, supported by a view of ethics that is ‘embodied, enacted and emergent’ (Adams et al. 2022, 6).

Adopting this ethical viewpoint necessitates moving beyond focusing on the ethical ramifications of algorithms alone (Beveridge, Figueiredo, and Holmes 2020). It calls for an appreciation of the complex interplay between humans and machines, particularly for the graduate student, who may find text generation tools impressive, produced at a speed and quality unimaginable (Schlagwein and Willcocks 2023). As graduate students interact with AI in writing tasks, their methods adjust to the capabilities and constraints of the technology. Writing studies have recognized this fluid negotiation with both human and non-human factors, suggesting a reimagined approach to developing ethical practices that emerge responsively within educational environments (Vetter et al. 2024).

4. Researcher positionality

My reflections in this conceptual paper are grounded in my current practice (Schön 2017), teaching on one Master’s of Education program in (name of country). The typical student demographics for this program are middle-aged, female, full-time teachers or school principals, wanting to complete a graduate degree for career advancement purposes. Particularly, this paper is situated within my experience of adopting an emergent pedagogy in teaching the educational research methodology course. While research ethics had always been a mainstay of this course, the advent of AI has reshuffled the priorities and focus areas within the syllabus, significantly changing when and how I approach teaching research ethics. In agreement with Smith (2016), I find the topic of research ethics inherently challenging to teach because it involves complex, often unresolved dilemmas rather than stable knowledge. Also, it is deeply influenced by context, as different academic disciplines and geographic locations adhere to a variety of legal, regulatory and cultural norms.

Situating my reflections within the field of education and the culturally dependent nature of research ethics, I used critical posthumanism as a theoretical lens to support

my assessment of students' entanglements with AI and their ability to produce a research proposal in English, as their non-native language. In addition to attending the lecture format of the course, students produced a research proposal, which gave them first-hand experience in implementing ethical practices. The aims of teaching research ethics through this course were to raise students' awareness of ethical issues in education, promote 'ethical sensibilities' (Smith 2016, 99) and engage students in ethical research activities. As part of a larger project on AI ethics in research contexts, this study received ethical approval from the IRB committee at (name of the university) and students consented to use their extracts from their research proposals. Below, I focus on three main tensions emerging from these entanglements, causing graduate students to struggle with disentangling themselves from AI as a co-author of text, while also considering the institutional and regulatory consequences arising from non-compliance with codes of conduct.

5. Emerging tensions

In this section, I explain three tensions that emerged from applying critical posthumanism, as a conceptual tool, to reflections on my practice of teaching an educational research methodology course. The tensions were termed: (1) blurring lines of authorship, (2) relinquishing agency, and (3) abating criticality. Each tension is explained below in reference to relevant scholarly work in the overlapping fields of AI, research ethics and academic writing.

5.1. *Blurring lines of authorship*

The fact is graduate students are more likely than not to experiment with and adapt generative AI (Vetter et al. 2024). Students taking the educational research methodology course admittedly acknowledged using ChatGPT and other AIs when questioned on a post-course survey – I could even sense their use of AI in responding to the survey questions. Scholars have recognized the benefits of AI on students' written products (Vetter et al. 2024), while also acknowledging its effects on the writing process. In exploring the performance of ChatGPT on writing essays, Robin (2023) observed the potential of AI in completing his assigned tasks with relative ease, concluding that students would be able to 'write' his assignments by managing the meaning-making process, yet without necessarily mastering the act of writing. He, among other writing instructors (AlAfnan et al. 2023), refrained from assigning take-home assignments, instead obliging students to complete writing tasks in class. While this may seem to work for short essays, graduate students expend a much longer time on devising their proposals or theses. Most of their writing is done unsupervised, and if they are to make the most of available tools at their disposal, then AI should be prioritized.

Disapproving the need to 'ban' AI from my research methodology course, I decided that students should learn how to use it responsibly and ethically. The first few sessions of the course were dedicated to these ethical discussions, including how-to instructions, university ethical guidelines and anecdotes from across the globe. However, as Smith (2016) contended, 'the lecture format was not conducive to exploration of these issues in depth', noting the constraints as 'encourage[ing] a transmission approach, giving

students little time to process or discuss issues or hear alternative views'. More importantly, the lecture format 'lacks the power to promote ethical sensibilities' (99). Drawing on the concept of assemblages, as explained by Leander and Burris (2020), I contended that research ethics were better taught as a multiplicity, emerging at the intersection of students' intra-actions with various AI tools, diverse ethical guidelines, course lectures and assignments, peers and instructor, among other personal elements (e.g. students' prior skills, experiences, commitments, etc.). This perspective, supported by the view that students' intra-actions with research ethics were emergent, relational and dynamic, approved students' entanglements with AI and led to blurring lines of authorship, which became an important topic for course discussions.

In addition to the lecture format, the course entailed written feedback on portions of the research proposal and online conference sessions with students as a means of discussing their progress, struggles and uncertainties. During these sessions, students conversed more consciously on their use of AI and the ethical dilemmas they faced, realizing (in some instances) the way that text-generating AI shaped their research ideas and writing. In retrospect, their persona as a novice researcher had not even developed, and arguably for many, was 'hijacked' by AI. Unlike an experienced researcher, who has engaged in research activities prior to AI, students' written proposals lack voice. The way they talked about their research during the conference sessions did not sound anything like they could have written on their own. English was not their native language, which may have amplified such blurring lines of authorship. According to Van Dis et al. (2023), while AI could help level the playing field by 'removing language barriers and enabling more people to write high-quality text', yet privileged researchers will always find ways to exploit AI in ways that accelerated their research and widened inequalities. On ethical grounds, removing these language barriers encouraged (perhaps 'innocent') violations of ethical guidelines, as students relinquished their agency to AI, as discussed in the next section.

5.2. Relinquishing agency

To what extent were students prepared to relinquish agency to AI? Obviously, a little too much. This excerpt from student A's research proposal shows who/what is enacting agency:

This study delves into the odyssey of educational leaders, tracing their path from first forays into the field to ascending the echelons of leadership, all while navigating the swift currents of technological advancement. It examines how leaders seamlessly integrate recent technologies into pedagogical and administrative frameworks, thereby reshaping their roles and pathways for career development. Moreover, this research scrutinizes the influence of educational policies and reforms on leadership roles, highlighting the hurdles and prospects that mold their professional journey. Through this lens, we aim to unearth not just the mechanisms through which leaders augment their ability and knowledge, but also their instrumental role in sculpting the educational future. This narrative investigation transcends a mere account of individual trajectories. It is an in-depth examination of how educational leadership serves as a cornerstone for forging resilient, forward-thinking educational entities adept at confronting the multifaceted challenges of the twenty-first century.

Drawing on critical posthumanism, I expected AI to act as a co-agent along with students in the writing of the research proposal. In fact, I encouraged this view of

distributive agency in both the lectures and the feedback sessions. I also expected a hybrid product, one ‘that is neither fully written by a human, nor by an AI’ (Eaton 2023, 3). However, unlike Eaton’s conviction, I *was* able to determine, quite readily, where the human ended and where the AI began. From the excerpt above, the human student ended with a prompt inserted into ChatGPT, and the AI took over from there. Human agency was completely suspended in this act, and the distributive quality of agency was abated.

A counterexample was also presented by a student in the research methodology course. In contrast to the excerpt above, student B also rejected the co-agency of AI and submitted this AI-free excerpt in the first draft of the research proposal:

The way leadership acts and behaves has a great influence on the behavior of students and the results they achieve. Research shows that successful leadership is has an important role in supporting and promoting students to obtain a great success. For example, a study made in Hong Kong spotted the light on the direct effect of primary leadership on school productivity, assuring the need for creating a strong leadership early among schools (Burton et al. 2005). Another study from the Philippines showed that leadership made a great transformation in learning environment and allowed students to have amazing results and satisfying performance (Estacio and Estacio 2022). These studies emphasize the importance of leadership in supporting and promoting students for a better education. However, there is a problem in understanding the ways leadership may affect and enhance student’s behavior in different educational fields. The aim of this study is to fill this void by investigating the between leadership practices and student behavior in school environments.

Agreeing with Vetter et al. (2024), student B seemed ‘to counter this idea that learning happens somewhere in between our sense of agency and other objects outside of ourselves’ (7), revealing contesting viewpoints among students sharing the same assemblage. The two excerpts represented extreme cases on opposite ends of an agentic continuum, both rejecting co-agency. Striking a balance between these ends and meeting distributive agency in midpoint was a challenge to the emergent pedagogy adopted in this course. Based on the need to (re)address ethical considerations, I dedicated an ensuing lecture to the fluidity of AI generated text and the responsibilities bestowed upon agentic students to revise, edit and adapt to the possibilities and limitations of the tool (Vetter et al. 2024). Furthermore, the online discussions opened the opportunity for students A and B, as well as other students like them, to negotiate instances of their AI (non-)use against the backdrop of what is permissible, responsible and ethical. Ongoing debates in many HE institutions on the use of AI generated text did not provide students with simple answers (Kramm and McKenna 2023). More importantly, an emergent goal was to engage students in thinking critically about AI outputs and weigh them against their readings of the literature, a theme discussed in the next section.

5.3. Abating criticality

The case of student A confirms the validity of the concerns in the literature ‘about the potential loss of student criticality in engaging these generative AI tools’ (Vetter et al. 2024, 8). Abating criticality, in this case, has meant the complete trust in the AI output and a possible ignorance of the technical nature of AI tools as mere probability engines, rather than an authority on a topic. Without attributing student A’s action to

an extrinsic desire for high grades or negative attitudes toward learning (Lines 2016), I could only speculate why this student resorted to such an extreme case of abating criticality: It may have been a lack of self-efficacy in one's writing ability, a lack of research experience and/or misunderstanding of ethical guidelines. In reference to critical posthumanism, instructors should not look for single causations for students' actions, as multiplicities are always at play.

Thus, Luke's (2018) predictions on the challenge raised by digital culture could, in fact, be both 'one of skills or technological competence' *and* 'one of participation and ethics' (185). Believing that 'students can still practice criticality if guided to do so' (Vetter et al. 2024, 7), I saw an emergent need to delve more seriously into the technicalities of AI tools and prompt engineering, as well as diversify the sources of knowledge transmission by incorporating relevant YouTube videos and Coursera modules. The online conference sessions continued to provide a space for learning and feedback. Following an iterative process, I guided students through the ethical use of AI, which entailed identifying a justifiable purpose for using AI, learning about relevant AI tools that serve this purpose, selecting the most reliable AI for their purpose, verifying and reviewing the output produced, and documenting and acknowledging the use of AI. Through these attempts, the aim was to support students' understanding of research ethics as an ongoing process of engaging with the socio-material world, and where their research activities would become entangled with diverse elements in an expanding assemblage.

If, as one student wrote in the post-course survey: 'AI is not ethical and lacks creativity and it makes human lazy and dependable', then students should come to see AI not as a substitution to their labor, but an enhancement (Vetter et al. 2024). In this case, students will learn to engage in some form of critical thinking, ensuring the reliability of AI outputs, instead of resorting to widespread paper mills and ghostwriters (Lines 2016), as has been the case in the past. So, far from seeing students' use of AI 'unethically' as a new problem, and far from my intention to take a 'police-catch-punish' approach (Kramm and McKenna 2023, 2173), course discussions focused on how 'AI could be used to foster a transformative relationship to knowledge' (Kramm and McKenna 2023, 2176). As novice researchers, students were given the benefit of the doubt that their unethical use of AI was unintentional and that they may use AI inappropriately, before they learn to use it critically (Lines 2016). Refuting a discourse of punishment (Lines 2016), course feedback also focused on 'situating the development of students' writing and research practices as a learning process' and 'finding ways to clarify students' understandings of appropriate source use' (893), which continued to feed into their submitted assignments until the end of the course.

6. Discussion

The surge of academic discourse on text generative AI denotes growing recognition of its implications, specifically for research activities (Foltynek et al. 2023; Van Dis et al. 2023). There is no doubt that AI has opened new possibilities for graduate students required to submit research proposals and theses within time constraints. Far from a computational algorithm belonging to computer science (Beveridge, Figueiredo, and Holmes 2020), AI is ethically consequential for graduate students. In effect, teaching of research ethics is

challenging, involving complex, often unresolved dilemmas rather than stable knowledge (Muthana, Chaaban, and Qadhi 2024; Smith 2016). Drawing on critical posthumanism as a theoretical lens (Strom and Martin 2022), this conceptual paper aimed to understand the tensions that arise through graduate students' entanglements and intra-actions with AI in the assemblage of an educational research methodology course. This work also explored ways to better support graduate students' development of locally emergent AI ethics through relevant pedagogies and policies.

Understanding the dynamic and emergent nature of human and nonhuman intra-actions carries implications for pedagogical considerations. As adopted in this course, emergent pedagogy, according to Rautio, Spector, and Thiel (2019), centers the unplanned 'disturbances' or 'distractions' as key sites of learning, rather than pre-designed plans (e.g. the lecture format). An emergent pedagogy is not merely about the interactions among human actors, but rather occurs through socio-material intra-actions, which further necessitates a rethinking of assessment procedures (Vetter et al. 2024). A major tenet of this course were pre-planned lectures, yet written feedback on portions of students' research proposals and online conference sessions as a means of discussing their progress, struggles and uncertainties were part of the emergent pedagogy adopted in this course and constitutes a main focus of this paper.

Through reflections on practice (Schön 2017), I identified three tensions that challenged the very notions of distributive agency and ethical criticality, influenced by students' intra-actions with AI, ethical guidelines, peers, instructor and other elements of the (course) assemblage. These tensions constituted blurring lines of authorship, relinquishing agency and abating criticality that challenged the development of a local AI ethics. Building on Vetter et al. (2024), the notion of a local ethics emerged in response to these tensions and through negotiations within the course assemblage. One observation in this regard was the need for students to have some form of computational/technical know-how, such that knowing how AI functions may prompt students to intra-act more critically with the AI, responding to critical posthumanism concerns of distributive agency and co-creations. Though an emergent pedagogy encouraged quick responses to students' needs, in retrospect, other forms of collaborative and multidisciplinary partnerships with colleagues in the computer science department may be considered for the future.

Another important observation pertained to the language barrier issue. While text generative AI can be useful in supporting non-native English researchers (Hwang et al. 2023), overreliance on AI has resulted in research products that have stifled students' research identities and lacked their voice. As novice researchers, students require additional time and effort to process and engage in research activities. Along with the benefits of AI, students will need to learn how to navigate the use of AI for language enhancement while preserving their persona as a researcher. Ongoing debates in many HE institutions on how to strike this balance have not been resolved (Kramm and McKenna 2023), requiring empirical investigations documenting the experiences of graduate students in their development of local ethics (Vetter et al. 2024).

In this research methodology course, students were not penalized for using AI in unethical ways; to the contrary, these instances were considered valuable learning opportunities for an emerging local ethics. In accordance with Smith (2016), the teaching of research ethics aimed to develop 'an ethical sensibility that is actively engaged by each

individual throughout all stages of the research process' (97). Despite the prolific emergence of ethical AI guidelines, there were indeed few mechanisms for compliance (Larsson 2020; Jang, Choi, and Kim 2022) and a mere reading of ethical guidelines was deemed insufficient (Smith 2016). Recognizing that these codes of conduct are pre-scriptive and static (Haggerty 2004), students were encouraged to scrutinize their use of AI and develop ethical criticality through their experiences. This approach addressed the tension between ethical compliance and ethical doings (Haggerty 2004), which remain relevant concepts with the advent of AI. Rather than relying exclusively on ethical guidelines, the course emphasized an iterative process where students identified a purpose for using AI, explored relevant AI tools, selected the most reliable option, verified and reviewed the output produced, and documented and acknowledged their use of AI.

Recent critiques of AI ethics have similarly questioned the efficacy of universal ethical guidelines, advocating instead for policies that recognize the fluidity of authorship and agency, and prepare students to navigate the ethical complexities of diverse discursive situations (Foltynek et al. 2023). This developmental perspective on ethics refutes a 'single universal ethics for AI usage', and seeks to 'move the field towards a framework for local negotiation and investigation, one that is also responsive to institutional and disciplinary guidelines' (Vetter et al. 2024, 3). Accordingly, the overreliance on universal and normative ethics prevents students from engaging critically with ethical tensions (Smith 2016), perhaps exacerbating the tensions of blurring lines of authorship, relinquishing agency and abating criticality. As policy mandates are dictated top-down, there is a need for students to engage in spaces where they can negotiate and deliberate ethical codes, without seeing them as completely pre-determined by university policy or course syllabi. Coming face to face with such ethical tensions, particularly in a safe environment where grades are not at stake, can support students in developing the kind of criticality needed to make ethical decisions not only for the known challenges of today, but for those that they will face in the future (Smith 2016). Finally, the realities of AI agents, including their evolving nature and pervasive use, mean that universities will need to update their policies at an unprecedented pace. Rather than wait for lengthy bureaucratic procedures to crystallize, students can be guided to move beyond compliance-driven approaches to acquiring ethical criticality, which combines awareness, knowledge, skill and commitment to AI ethics.

7. Conclusion

In this conceptual paper, I have argued that graduate students grapple with the use of AI in their research activities in ethical and responsible ways. While students may easily be accused of cheating or plagiarism, taking a critical posthumanism perspective revealed an alternative reality; one where several tensions arise and need to be addressed along the boundaries of ethics, research and AI. This paper contributes to a theoretical understanding of what it means for graduate students to do research in the age of AI, pointing to the need for further empirical studies on the effectiveness of emergent pedagogy, teaching with uncertainty and multidisciplinary instruction. Given the relatively recent interest in AI-enhanced research and scholarship, further empirical studies that delve into the experiences of graduate students are needed to better understand their intra-actions with locally constructed AI ethics. Pedagogically, enhancing ethical sensibilities and

modifying syllabi to accommodate critical posthumanism concepts may be readily achievable individual actions for anyone who teaches educational research methodology. However, addressing the policy challenges inherent in overly prescriptive codes of conduct and institutional bureaucracies may take a little longer.

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References

- Adams, C., P. Pente, G. Lermeyer, J. Turville, and G. Rockwell. 2022. Artificial intelligence and teachers' new ethical obligations. *The International Review of Information Ethics* 31, no. 1: 1–18. doi:10.29173/irie483.
- AlAfnan, M.A., S. Dishari, M. Jovic, and K. Lomidze. 2023. Chatgpt as an educational tool: Opportunities, challenges, and recommendations for communication, business writing, and composition courses. *Journal of Artificial Intelligence and Technology* 3, no. 2: 60–8. doi:10.37965/jait.2023.0184.
- Barad, K. 2007. *Meeting the universe halfway: Quantum physics and the entanglement of matter and meaning*. Durham, NC: Duke University Press.
- Bennett, J. 2010. *Vibrant matter: A political ecology of things*. Durham, NC: Duke University Press.
- Beveridge, A., S.C. Figueiredo, and S. Holmes. 2020. Introduction to composing algorithms: Writing (with) rhetorical machines. *Computers and Composition* 57: 102594. doi:10.1016/j.compcom.2020.102594.
- Braidotti, R. 2013. *The posthuman*. Cambridge, UK: Polity Press.
- Braidotti, R. 2019. A theoretical framework for the critical posthumanities. *Theory, Culture and Society* 36, no. 6: 31–61. doi:10.1177/0263276418771486.
- Chaaban, Y., S. Qadhi, J. Chen, and X. Du. 2024. Understanding Researchers' AI readiness in a higher education context: Q methodology research. *Education Sciences* 14, no. 7: 709. doi:10.3390/educsci14070709.
- Coffey, L. 2023. AI and peer review: Enemies or allies? Retrieved 20 February, 2024 from: <https://www.insidehighered.com/news/tech-innovation/artificial-intelligence/2023/10/24/ai-can-lesser-peer-review-woes-researchers>.
- Conroy, G. 2023. How ChatGPT and other AI tools could disrupt scientific publishing. *Nature* 622: 234–6. doi:10.1038/d41586-023-03144-w.
- Deleuze, G., and F. Guattari. 1987. *Capitalism and schizophrenia: A thousand plateaus*. Minneapolis: University of Minnesota Press.
- Eaton, S.E. 2023. Postplagiarism: Transdisciplinary ethics and integrity in the age of artificial intelligence and neurotechnology. *International Journal for Educational Integrity* 19, no. 1: 23. doi:10.1007/s40979-023-00144-1.

- Foltynek, T., S. Bjelobaba, I. Glendinning, Z.R. Khan, R. Santos, P. Pavletic, and J. Kravjar. 2023. ENAI recommendations on the ethical use of Artificial Intelligence in education. *International Journal for Educational Integrity* 19, no. 1: 1–4. doi:[10.1007/s40979-023-00133-4](https://doi.org/10.1007/s40979-023-00133-4).
- Frisicaró-Pawlowski, E. 2018. Rhetorical ethics and the language of virtue: Problems of agency and action. *College English* 81, no. 2: 110–32. doi:[10.58680/ce201829859](https://doi.org/10.58680/ce201829859).
- Haggerty, K.D. 2004. Ethics creep: Governing social science research in the name of ethics. *Qualitative sociology* 27: 391–414.
- Hill-Yardin, E.L., M.R. Hutchinson, R. Laycock, and S.J. Spencer. 2023. A Chat (GPT) about the future of scientific publishing. *Brain, Behavior, and Immunity* 110: 152–4. doi:[10.1016/j.bbi.2023.02.022](https://doi.org/10.1016/j.bbi.2023.02.022).
- Hwang, S.I., J.S. Lim, R.W. Lee, Y. Matsui, T. Iguchi, T. Hiraki, and H. Ahn. 2023. Is ChatGPT a “fire of prometheus” for non-native English-speaking researchers in academic writing? *Korean Journal of Radiology* 24, no. 10: 952. doi:[10.3348/kjr.2023.0773](https://doi.org/10.3348/kjr.2023.0773).
- Jang, Y., S. Choi, and H. Kim. 2022. Development and validation of an instrument to measure undergraduate students’ attitudes toward the ethics of artificial intelligence (AT-EAI) and analysis of its difference by gender and experience of AI education. *Education and Information Technologies* 27, no. 8: 11635–67. doi:[10.1007/s10639-022-11086-5](https://doi.org/10.1007/s10639-022-11086-5).
- Johnstone, J. 2007. Technology as empowerment: A capability approach to computer ethics. *Ethics and Information Technology* 9, no. 1: 73–87. doi:[10.1007/s10676-006-9127-x](https://doi.org/10.1007/s10676-006-9127-x).
- Kramm, N., and S. McKenna. 2023. AI amplifies the tough question: What is higher education really for? *Teaching in Higher Education* 28, no. 8: 2173–8. doi:[10.1080/13562517.2023.2263839](https://doi.org/10.1080/13562517.2023.2263839).
- Larsson, S. 2020. On the governance of artificial intelligence through ethics guidelines. *Asian Journal of Law and Society* 7, no. 3: 437–51. doi:[10.1017/als.2020.19](https://doi.org/10.1017/als.2020.19).
- Latour, B. 2005. *Reassembling the social: An introduction to Actor-Network-Theory*. Oxford: Oxford University Press.
- Leander, K.M., and S.K. Burriss. 2020. Critical literacy for a posthuman world: When people read, and become, with machines. *British Journal of Educational Technology* 51, no. 4: 1262–76. doi:[10.1111/bjet.12924](https://doi.org/10.1111/bjet.12924).
- Lines, L. 2016. Ghostwriters guaranteeing grades? The quality of online ghostwriting services available to tertiary students in Australia. *Teaching in Higher Education* 21, no. 8: 889–914. doi:[10.1080/13562517.2016.1198759](https://doi.org/10.1080/13562517.2016.1198759).
- Luke, A. 2018. Digital ethics now. *Language and Literacy* 20, no. 3: 185–98. doi:[10.20360/langandlit29416](https://doi.org/10.20360/langandlit29416).
- Miller, E.R. 2016. The Ideology of Learner Agency and the Neoliberal Self. *International Journal of Applied Linguistics* 26, no. 3: 348–365. doi:[10.1111/ijal.12129](https://doi.org/10.1111/ijal.12129).
- Morton, T. 2013. *Hyperobjects: Philosophy and ecology after the end of the world*. Minneapolis: University of Minnesota Press.
- Muthana, A., Y. Chaaban, and S. Qadhi. 2024. A model of the interrelationship between research ethics and research integrity. *International Journal of Qualitative Studies on Health and Well-being* 19: 2295151. doi:[10.1080/17482631.2023.2295151](https://doi.org/10.1080/17482631.2023.2295151).
- Nath, R., and R. Manna. 2023. From posthumanism to ethics of artificial intelligence. *AI and Society* 38, no. 1: 185–96. doi:[10.1007/s00146-021-01274-1](https://doi.org/10.1007/s00146-021-01274-1).
- Rautio, P., K. Spector, and J.J. Thiel. 2019. Theory that cats have about swift louseflies: A distractive response. In *Posthumanism and literacy education: Knowing/becoming/doing literacies*, edited by C. R. Kuby, K. Spector, and J. J. Thiel, 228–234. Routledge.
- Robin, C. 2023. The end of the take home essay? How ChatGPT changed my plans for the fall. *The Chronicle*. August 24. https://www.chronicle.com/article/the-end-of-the-take-home-essay?cid=gen_sign_inandsra=true.
- Schlagwein, D., and L. Willcocks. 2023. ‘ChatGPT et al.’: The ethics of using (generative) artificial intelligence in research and science. *Journal of Information Technology* 38, no. 3: 232–8. doi:[10.1177/02683962231200411](https://doi.org/10.1177/02683962231200411).
- Schön, D.A. 2017. *The reflective practitioner: How professionals think in action*. London: Routledge.

- Smith, J. 2016. Reflections on teaching research ethics in education for international postgraduate students in the UK. *Teaching in Higher Education* 21, no. 1: 94–105. doi:[10.1080/13562517.2015.1115968](https://doi.org/10.1080/13562517.2015.1115968).
- Strom, K.J., and A.D. Martin. 2022. Toward a critical posthuman understanding of teacher development and practice: A multi-case study of beginning teachers. *Teaching and Teacher Education* 114: 103688. doi:[10.1016/j.tate.2022.103688](https://doi.org/10.1016/j.tate.2022.103688).
- Taylor, C.A., and A. Bayley. 2019. *Posthumanism and higher education: Reimagining pedagogy, practice and research*. London: Springer.
- UNESCO. 2023. *Guidance for generative AI in education and research*. Paris: United Nations Educational, Scientific and Cultural Organization. <https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research>
- Van Dis, E.A., J. Bollen, W. Zuidema, R. Van Rooij, and C.L. Bockting. 2023. ChatGPT: Five priorities for research. *Nature* 614, no. 7947: 224–6. doi:[10.1038/d41586-023-00288-7](https://doi.org/10.1038/d41586-023-00288-7).
- Van Noorden, R., and J.M. Perkel. 2023. AI and science: what 1,600 researchers think. *Nature* 621, no. 7980: 672–675. doi:[10.1038/d41586-023-02980-0](https://doi.org/10.1038/d41586-023-02980-0).
- Vetter, M.A., B. Lucia, J. Jiang, and M. Othman. 2024. Towards a framework for local interrogation of AI ethics: A case study on text generators, academic integrity, and composing with ChatGPT. *Computers and Composition* 71: 102831. <https://doi.org/10.1016/j.compcom.2024.102831>.